

Maharashtra Public Service Commission Assessment

Subject Name :	Associate Professor Instrumentation Engg Group A
Duration :	60
Number of Questions :	100
Total Marks :	200
Display Marks:	Yes

Maximum Instruction Time :	0
Is Section Default? :	No

Question Number : 1 Question Id : 464662601 Question Type : MCQ

Particular Integral (P.I.) of the differential equation $(D^2 + 5D + 6)y = e^x$ is

Options :

1. e^x

2. $\frac{e^x}{12}$

3. $c_1 e^{2x} + c_2 e^{3x}$

4. $c_1 e^{-2x} + c_2 e^{-3x}$

If λ is an eigen value of a matrix A then $\frac{1}{\lambda}$ is the eigen value of

Options :

1. transpose of matrix A
2. inverse of matrix A
3. matrix A
4. matrix A^2

The value of the line integral $I = \int_C 3x^2 dx + 2yz dy + y^2 dz$, where C is a curve joining the points $(0, 1, 2)$ and $(1, -1, 7)$ is

Options :

1. 0

2. 3

3. 4

4. 6

What is the particular solution of the differential equation $\frac{dy}{y} = 3x^2 dx$ satisfying the condition that $y = 2$ at $x = 0$?

Options :

1. $3e^{x^2}$

2. $3e^{x^3}$

3. $2e^{x^3}$

4. $2 \log_e |x|$

Question Number : 5 Question Id : 464662605 Question Type : MCQ

Let G be a region and suppose that f is a non-constant analytic function on G .
If U is an open set in G then $f(U)$ is

Options :

1. bounded
2. closed
3. unbounded
4. open

Question Number : 6 Question Id : 464662606 Question Type : MCQ

Out of 2000 families with 4 children each, the number of families you would expect to have no girls is,

$p = \text{probability of having a boy} = 0.5$

$q = \text{probability of having a girl} = 0.5$

Options :

1. 300
2. 150
3. 200
4. 125

The value of the integral $I = \int_C \frac{z}{z+1} dz$, where, C being given by $|z| = 2$ is

Options :

1. 1
2. -1
3. $2\pi i$
4. $-2\pi i$

Find the mean of the following data.

Numbers	:	8	10	15	20
Frequency	:	5	8	8	4

Options :

1. $\frac{320}{25}$
2. 320
3. 25
4. 08

Question Number : 9 Question Id : 464662609 Question Type : MCQ

A coil has a self inductance of 0.1 Henry. If the current in the coil is changed from 10 A to -10 A in 0.5 seconds, the magnitude of emf induced in the coil will be

Options :

1. 10 V
2. 2 V
3. 4 V
4. Zero

Energy required to overcome the opposition of self-induced emf during the current growth is

Options :

1. Dissipated as heat
2. Stored in electric field
3. Stored in magnetic field
4. Lost due to resistance only

Which statement best explains, why magnetic flux always forms closed loops even in non-magnetic media ?

Options :

1. Magnetic monopoles do not exists
2. Permeability of air is zero
3. Magnetic field lines originate only from current
4. Flux closure is due to electrical conductivity

Which of the following expressions is incorrect ?

Options :

1.
$$C = \frac{\epsilon d}{A}$$

2.
$$C = \frac{Q}{V}$$

3.
$$Q = \int i dt$$

4.
$$C = \frac{\epsilon A}{d}$$

“The outflow of flux from a small closed surface per unit volume, as the volume shrinks to zero”, is called

Options :

1. The divergence of vector flux density
2. The curl of vector flux density
3. The electric flux density
4. The electric field intensity

Question Number : 14 Question Id : 464662614 Question Type : MCQ

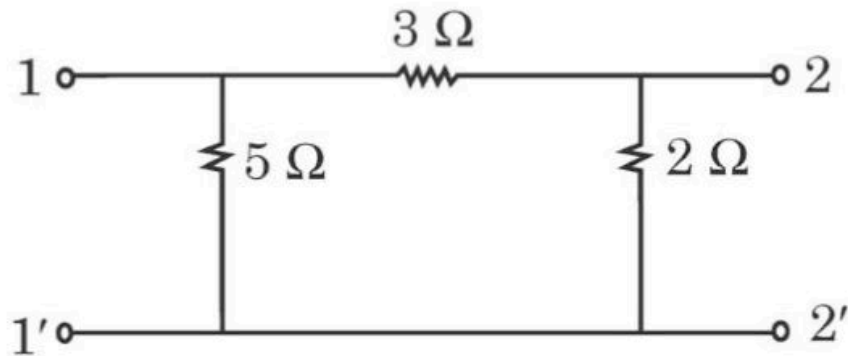
The negative sign in Faraday's law for the time varying magnetic field represents

Options :

1. Direction of current flow
2. Energy conservation
3. Lenz's law
4. Magnetic permeability

Question Number : 15 Question Id : 464662615 Question Type : MCQ

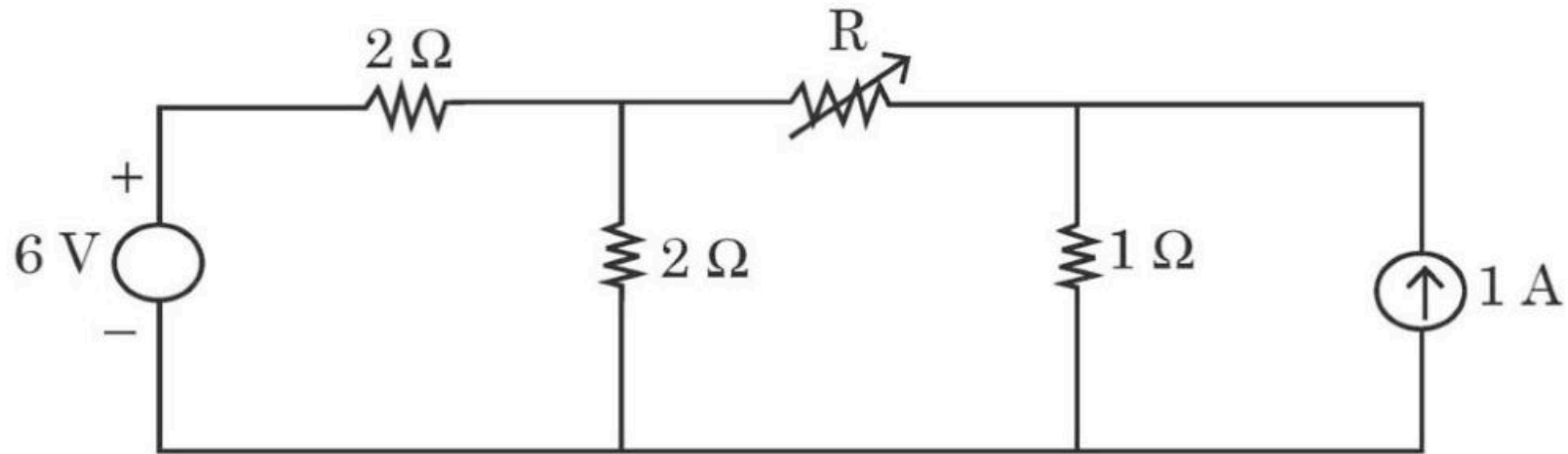
In the network given below, the value of open circuit forward transfer impedance (Z_{21} parameter) will be



Options :

1. 1 Ω
2. 2 Ω
3. 3 Ω
4. 4 Ω

Consider the following network



The value of maximum power, that can be absorbed by resistance R , is

Options :

1. 0.25 W
2. 0.5 W
3. 1 W
4. 4 W

Question Number : 17 Question Id : 464662617 Question Type : MCQ

A sinusoidal voltage $v = 50 \sin(\omega t)$ is applied to a series R-L circuit. The current in the circuit is given by $i = 20 \sin\left(\omega t - \frac{\pi}{3}\right)$. What will be the value of active power ?

Options :

1. 1000 W
2. 500 W
3. 250 W
4. 150 W

In a short dst. test, the applied voltage is typically (5 – 10)% of rated voltage because,

Options :

1. Rated voltage would cause excessive copper loss
2. Low voltage ensures rated current without core saturation
3. Iron loss becomes dominant at low voltage
4. Leakage flux become negligible

For a three-phase induction motor having rotor circuit resistance of $6\ \Omega$, maximum torque occurs at a slip of 0.6. The value of standstill rotor circuit reactance is

Options :

1. $4.44\ \Omega$
2. $0.36\ \Omega$
3. $1\ \Omega$
4. $10\ \Omega$

Superposition theorem is valid for

Options :

1. Linear circuits
2. Non-linear circuits
3. Both linear and non-linear circuits
4. Neither for linear nor for non-linear circuits

Question Number : 21 Question Id : 464662621 Question Type : MCQ

A voltage $v = 100 \sin(314t - 30^\circ)$ is connected across a 10Ω resistor. The power dissipated in the circuit will be

Options :

1. 1000 W
2. 500 W
3. 250 W
4. None of these

Question Number : 22 Question Id : 464662622 Question Type : MCQ

During short circuit test at full-load current the power input to a transformer comprises predominantly

Options :

1. Eddy current loss
2. Copper loss
3. Core loss
4. Both copper loss and core loss

Unit impulse sequence $\delta(n)$ in terms of unit step sequence is given by

Options :

1. $u(n) + u(n + 1)$

2. $u(n) - u(n - 1)$

3. $u(n) + u(n - 1)$

4. $u(n) - u(n + 1)$

Why is it important to consider the Region of Convergence (ROC) of $X(s)$ when defining the Laplace transforms ?

Options :

1.

Because the Laplace transform integral only converges for certain values of the complex variables

2.

Because ROC determines whether the system is discrete-time or continuous-time

3.

Because ROC tells us whether the time-domain signal is periodic or aperiodic

4.

Because ROC defines only the imaginary part of s , i.e. ω

Energy density spectrum of a signal $x(n) = a^n u(n)$ is

Options :

1.
$$\frac{1}{1 - 2a \cos \omega + a^2}$$

2.
$$\frac{1}{1 - 5a \cos \omega + a^2}$$

3.
$$\frac{1}{1 + 2a \cos \omega + a^2}$$

4.
$$\frac{1}{1 + 5a \cos \omega + a^2}$$

The filter with the system function $H(z) = z^{-k}$ is a

Options :

1. Notch filter
2. Band pass filter
3. All pass filter
4. Low pass filter

The z transform of the sequence $x(n) = a^n u(n)$ is

Options :

1.
$$X(z) = \frac{1}{1 - az^2}$$

2.
$$X(z) = \frac{1}{1 - az^{-1}}$$

3.
$$X(z) = \frac{1}{1 + az^2}$$

4.
$$X(z) = \frac{1}{1 + az^{-1}}$$

The Laplace transform of a function $\sinh(\omega t)$

Options :

1.
$$\frac{\omega}{s^2 + \omega^2}$$

2.
$$\frac{s}{s^2 + \omega^2}$$

3.
$$\frac{1}{s^2 - \omega^2}$$

4.
$$\frac{\omega}{s^2 - \omega^2}$$

Region of convergence for $h(n) = -\left(\frac{1}{2}\right)^n u(-n-1)$

Options :

1. $|z| < \frac{1}{2}$

2. $|z| > \frac{1}{2}$

3. $0 < |z| < \infty$

4. $|z| < \infty$

What is the sequence $y(n)$ that results from the use of four point DFTs if the impulse response is $h(n) = \{1, 2, 3\}$ and the input sequence is $x(n) = \{1, 2, 2, 1\}$?

Options :

1. $\{9, 9, 7, 11\}$
2. $\{1, 4, 9, 11, 8, 3\}$
3. $\{7, 9, 7, 11\}$
4. $\{9, 7, 9, 11\}$

Find the convolution $x(n) * h(n)$ for

$$x(n) = \{3, 1, 2, -1\}$$

$\uparrow$

$$h(n) = \{3, 2, 1\}$$

$\uparrow$

Options :

1. $\{9, 11, 9, 2, 0, -1\}$

$\uparrow$

2. $\{-1, 0, 2, 9, 11, 9\}$

$\uparrow$

3. $\{9, 9, 11, 2, 0, -1\}$

$\uparrow$

4. $\{9, 11, 9, 2, 0, -1\}$

$\uparrow$

The system described by a equation $Y[n] = \frac{1}{3}[x(n+1) + x(n) + x(n-1)]$ is

Options :

1. High pass filter
2. Moving average filter
3. Band pass filter
4. Band stop filter

The Laplace transform of a $x(t) = e^{2t} u(t) + e^{-3t} u(-t)$ is

Options :

1.
$$\frac{5}{s^2 + s - 6}$$

2.
$$\frac{5}{s^2 + s + 6}$$

3.
$$\frac{1}{s^2 + s - 6}$$

4.
$$\frac{1}{s^2 + s + 6}$$

Which of the following is a Dirichlet condition with respect to the signal $X(t)$?

Options :

1.
Signal $X(t)$ has a finite number of discontinuities in any period
2.
Signal $X(t)$ has a finite number of maxima and minima during any period
3.
Signal $X(t)$ is absolutely integrable in any period
4.
All of the above

Question Number : 35 Question Id : 464662635 Question Type : MCQ

Three blocks with gains 4, 8 and 10 are connected in parallel. The total gain of this arrangement is

Options :

1. 84

2. 42

3. 320

4. 22

The compensated system has a pole at the origin. What is the type of such a system ?

Options :

1. Type 0
2. Type 1
3. Type 2
4. Undetermined

Voltage in (V) Electrical System is analogous to which variable in Liquid Level and thermal system ?

Options :

1. Liquid flow (m^3), Heat flow (I)
2. Capacitance (m^2), Capacitance ($\text{J}/^\circ\text{C}$)
3. Liquid flow rate (m^3/min), Heat flow rate (J/min)
4. Head (m), Temperature ($^\circ\text{C}$)

The open loop transfer function of a unity feedback system is

$$G(s) = \frac{K}{s(s+2)(s+4)}; \text{ the range of } K \text{ for a stability or stable operation is}$$

Options :

1. $0 < K < 48$

2. $0 < K < 24$

3. $0 < K < 36$

4. $0 < K < 42$

Question Number : 39 Question Id : 464662639 Question Type : MCQ

Area under a unit impulse function is

Options :

1. Infinity
2. Unity
3. Zero
4. Always negative value

Open loop transfer function of system

$$G(s) H(s) = \frac{K}{s(s+2)(s+4)(s+6)}$$

The valid breakaway points those lie in root locus regions are

Options :

1. $-3, -5$

2. $-5.263, -0.763$

3. $5.263, 0.763$

4. $-2, -4$

Minimum phase system are having

Options :

1. Only poles are lies in the RHS s-plane
2. Only zeros are lies in the RHS s-plane
3. Poles and zeros both lies in the RHS s-plane
4. Poles and zeros both lies in the LHS s-plane

Lead compensators improves

Options :

1. Settling time
2. Rise time
3. Peak time
4. All of the above

A unity feedback system is characterized by open loop transfer function

$$G(s) = \frac{1}{s(0.5s + 1)(0.2s + 1)}$$

Which is correct for position error constant (K_p) and velocity error constant (K_v) ?

Options :

1. $K_p = 0$ and $K_v = \infty$

2. $K_p = \infty$ and $K_v = 0$

3. $K_p = \infty$ and $K_v = 1$

4. $K_p = 1$ and $K_v = \infty$

A system transfer function is given by

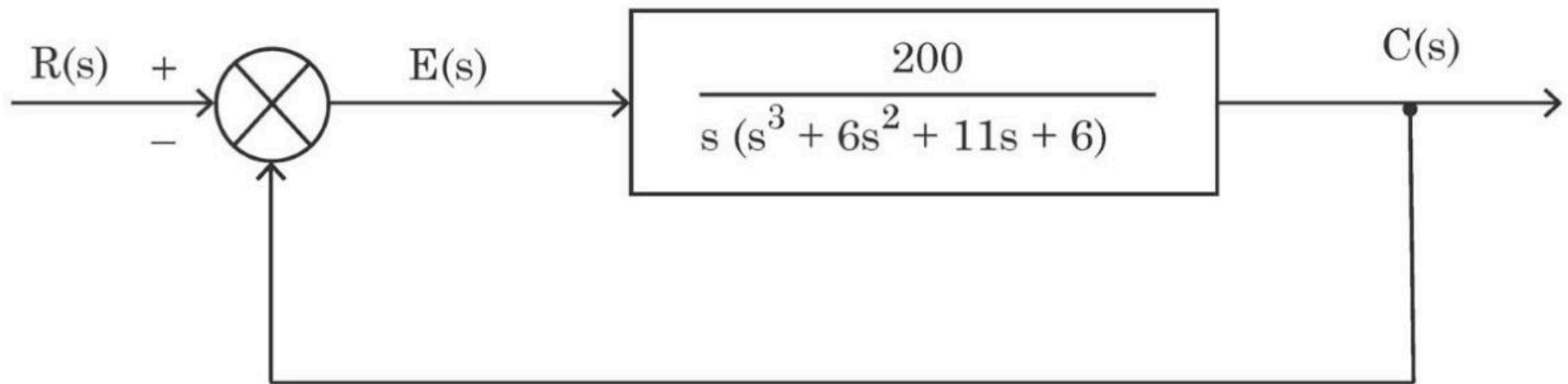
$$G(s) = \frac{K(s+3)}{s(s+1)(s+2)(s+4)}$$

The root locus will lie on the real axis

Options :

1. Between $s = 0$ and $s = -1$
2. Between $s = -1$ and $s = -2$
3. Between $s = 0$ and $s = 1$
4. Between $s = 1$ and $s = 2$

Consider the LTI system shown below



The given system is

Options :

1. Stable
2. Unstable
3. Marginally stable
4. None of the above

Fail state configuration of valve fail to open will _____.

- a. Valve is closed by applying pressure to pressure port
- b. If system fails, valve will revert back to open position
- c. Valve is opened by applying pressure to pressure port
- d. If system fails, valve will go to closed position

Options :

- 1. Option a and b is correct
- 2. Option c and d is correct
- 3. Option b and d is correct
- 4. Option a and c is correct

A closed-loop transfer function of a system is given as $T(s) = \frac{200}{s^4 + 6s^3 + 11s^2 + 6s + 200}$

The system has the number of poles in the left-half and right-half of s-plane are

Options :

1. Two right-half and two left-half of s-plane poles
2. Three right-half and one left-half of s-plane poles
3. One right-half and three left-half of s-plane poles
4. None of the above

The open loop transfer function of a unity feedback system is

$$G(s) = \frac{K}{s(1 + 0.1s)(1 + s)}$$

the value of K for the 6 dB GM is

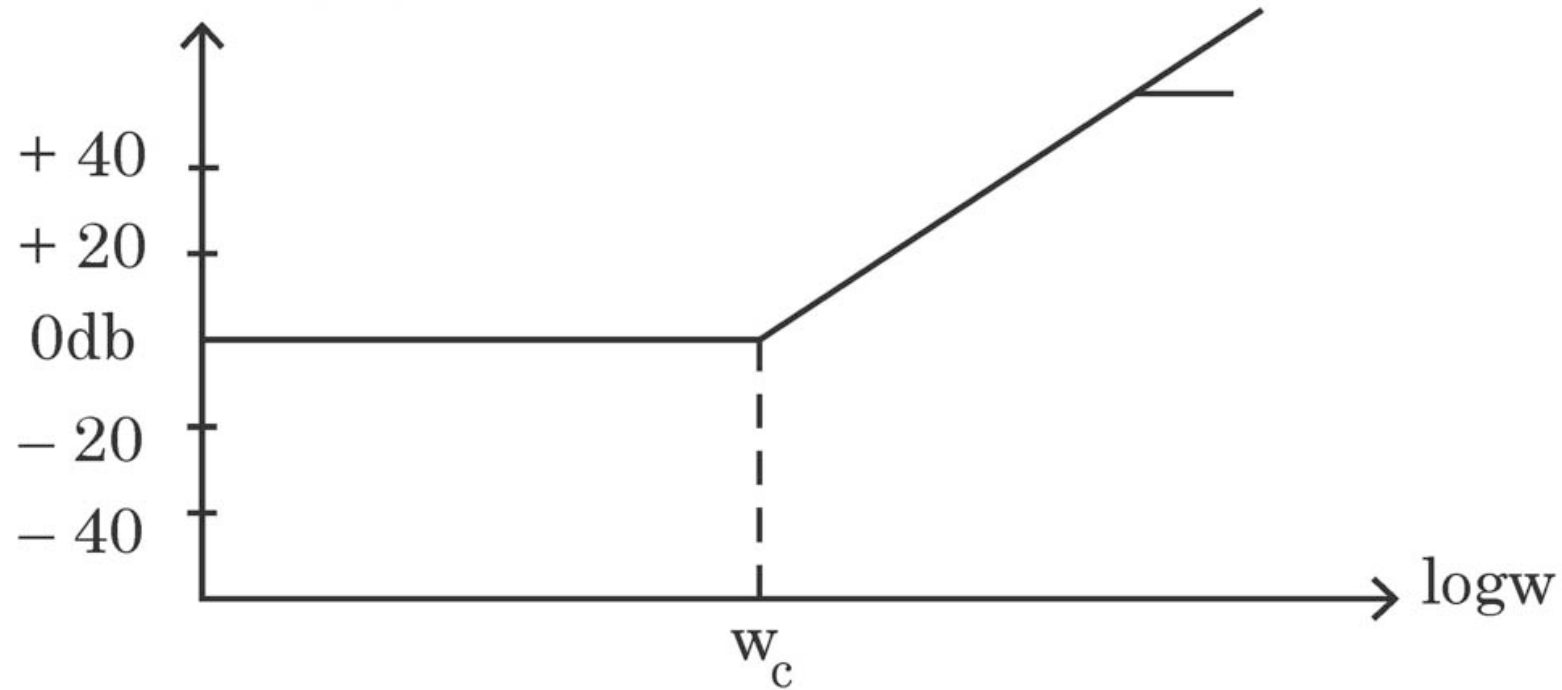
Options :

1. 4.5
2. 5.5
3. 1
4. 20

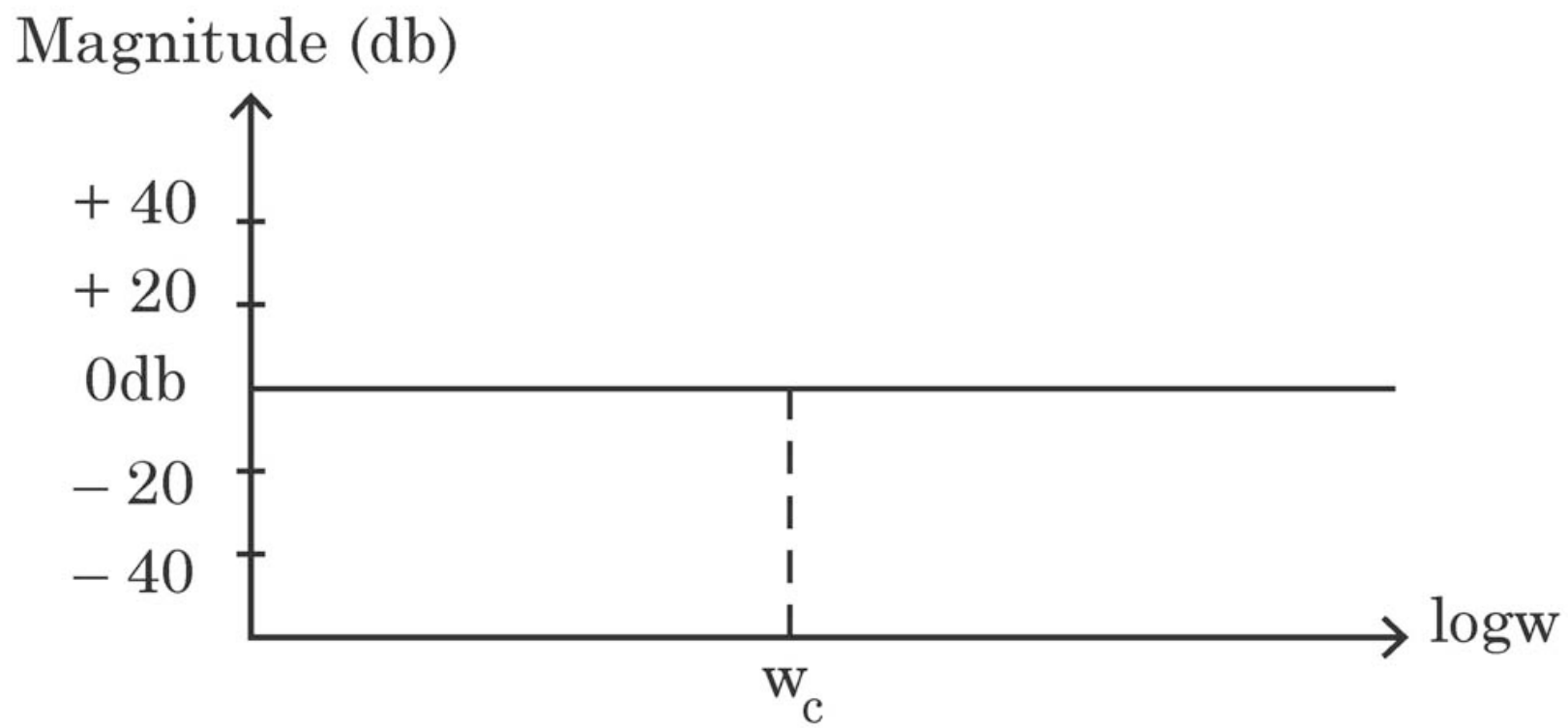
Bode magnitude plot for $G(s) = 1 + sT$ is given by

Options :

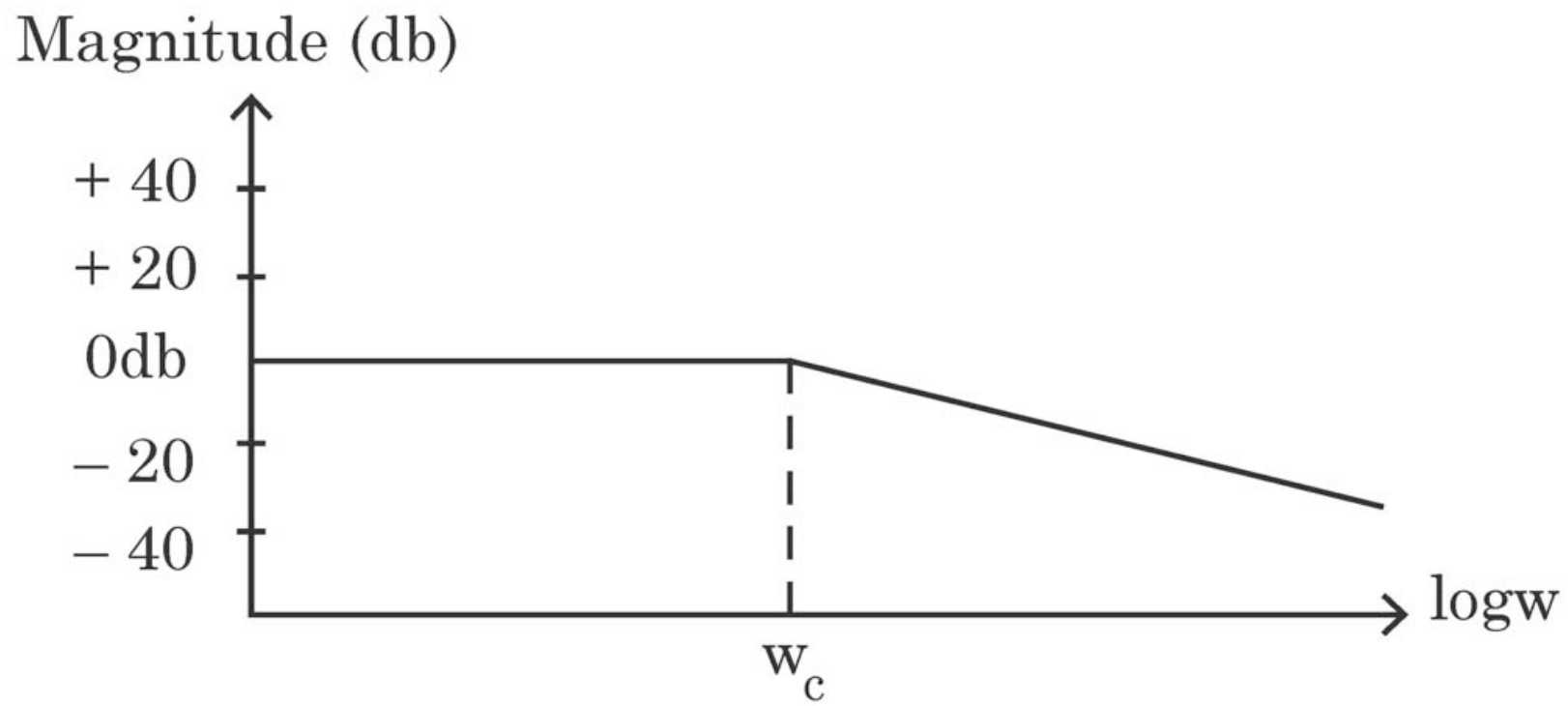
Magnitude (db)



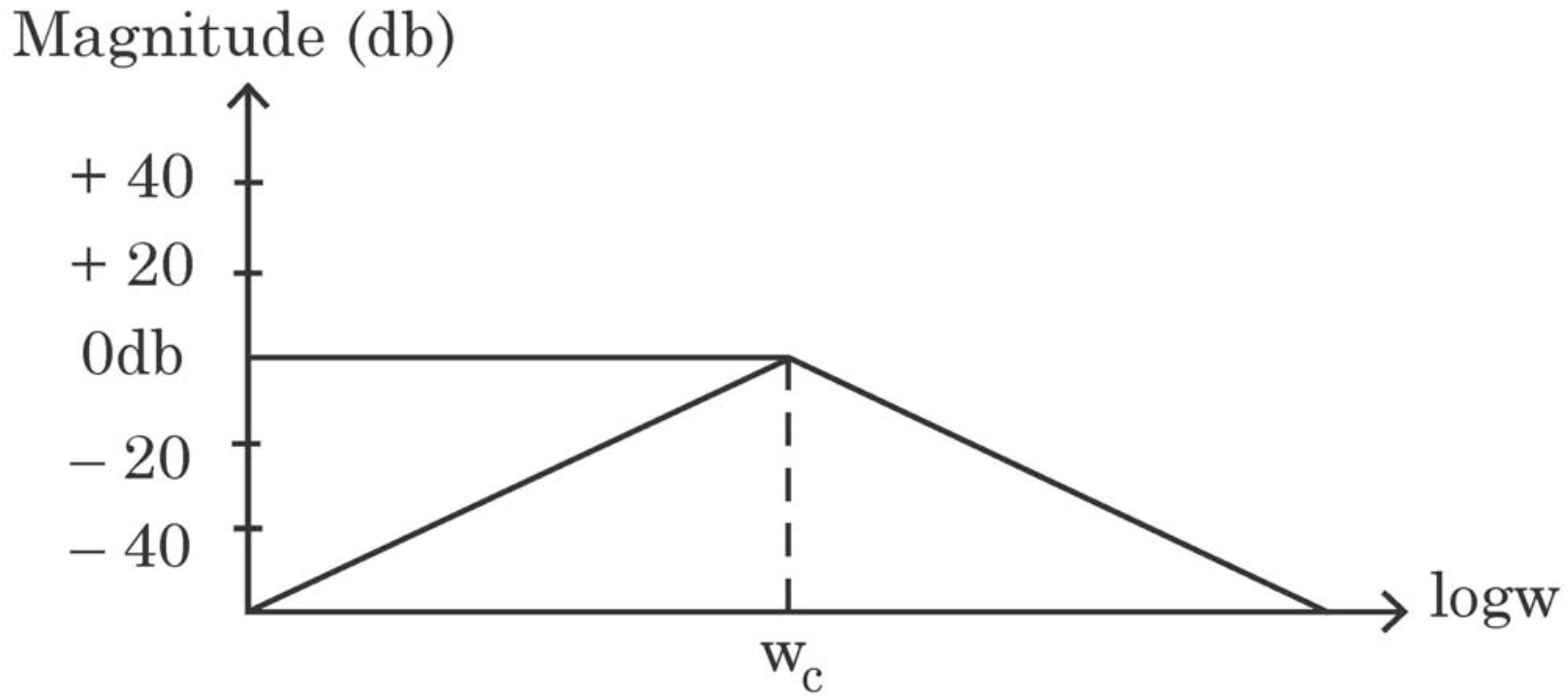
1.



2.



3.



4.

Question Number : 50 Question Id : 464662650 Question Type : MCQ

For Op-amp as an amplifier $V_o = A_v \cdot V_i$, A_v is voltage gain has a ideal value of

Options :

1. ∞

2. $-\infty$

3. 1000

4. -100

Op-amp specifications include

Options :

1. Offset voltage and currents
2. Frequency parameter, gain bandwidth
3. Slew rate
4. All of the above

Measurement of phase difference between two sinusoidal signals of the same frequency using CRO is

Options :

$$\sin^{-1} \left(\frac{x_1}{x_2} \right)$$

1.

$$\sin^{-1} \left(\frac{x_1}{y_1} \right)$$

2.

$$\sin^{-1} \left(\frac{y_2}{x_1} \right)$$

3.

$$\sin^{-1} \left(\frac{y_1}{x_2} \right)$$

4.

Question Number : 53 Question Id : 464662653 Question Type : MCQ

A certain Zener diode has a maximum power rating of 300 mW at 50°C and a derating factor of 3.2 mW/°C. Determine the maximum power the Zener diode can dissipate at a temperature of 90°C.

Options :

1. 272 mW
2. 72 mW
3. 172 mW
4. 372 mW

Question Number : 54 Question Id : 464662654 Question Type : MCQ

Common emitter amplifier is inverting so the output signal w.r.t. input signal is following degree out of phase

Options :

1. 360°
2. 90°
3. 180°
4. 270°

As in case of all types of oscillation circuits, the gain must be a minimum of _____ to ensure oscillations.

Options :

1. 2
2. 1
3. 10
4. 0.5

Question Number : 56 Question Id : 464662656 Question Type : MCQ

An operational amplifier has a bandwidth of 100 kHz. Negative feedback is introduced with a loop gain of 50. The bandwidth with feedback is

Options :

1. 1.96 kHz
2. 2 kHz
3. 49 kHz
4. 5.1 MHz

Which of the following statements about FET and BJT are correct ?

A : FET's are more temperature stable than BJT's

B : FET's are usually smaller than BJT's making them particularly useful in integrated circuits chips.

Choose the correct option.

Options :

1. Statement A is correct and statement B is incorrect
2. Statement A is incorrect and statement B is correct
3. Both statements are incorrect
4. Both statements are correct

Which of the following best describes a combinational logic circuit ?

Options :

1. Its output depends only on past inputs stored in memory elements
2. Its output depends on both present inputs and stored states
3. Its output depends only on the present input values
4. Its output repeats periodically regardless of input

A MUX with its address bits generated by a counter operates as a

Options :

1. Parallel to serial converter
2. Serial to parallel converter
3. Modified counter
4. Modified multiplexer

De Morgan's law states that

Options :

1.
$$\left(\overline{A + B}\right) = \overline{A} + \overline{B}$$

2.
$$\left(\overline{AB}\right) = A + B$$

3.
$$\left(\overline{AB}\right) = \overline{A} \cdot \overline{B}$$

4.
$$\left(\overline{A + B}\right) = \overline{A} \cdot \overline{B}$$

Which of the following Boolean algebraic expression is incorrect ?

Options :

1. $(A + B)(A + C) = A + BC$

2. $A + \overline{A}B = A + B$

3. $(A + \overline{B})(A + B) = A$

4. $A + AB = B$

Question Number : 62 Question Id : 464662662 Question Type : MCQ

A successive approximation A/D converter has a resolution 20 mV. What will be its digital output for an analog input of 2.17 V ?

Options :

1. 0110 1100
2. 0110 1101
3. 0110 1011
4. 0110 1110

Convert $(247.36)_8$ to equivalent hexa decimal number

Options :

1. $(A7.78)_{16}$

2. $(7A.87)_{16}$

3. $(B7.78)_{16}$

4. $(AB.78)_{16}$

It is desired to mask (set to 0) the most significant four bits of a byte in the accumulator. Suggest an instruction to accomplish this in an 8085 μ p system.

Options :

1. ANI FOH
2. CMP A
3. ANI OFH
4. ADI FOH

In LSI, the number of gate circuits per chip is

Options :

1. 100 to 999
2. 100 to 9,999
3. 1,000 to 10,000
4. $< 10,000$

Which of the following is not a base unit in the SI (International System of Units) ?

- A. Second
- B. Mole
- C. Lumen
- D. Ampere

Options :

1. A and B

2. Only A

3. Only C

4. D and C

The “Probable Error (P.E.)” in a Gaussian distributed set of readings is defined as

Options :

1. The standard deviation multiplied by 1.96
2. The value r such that 50% of readings lie within $\pm r$ of the mean
3. Four times the standard deviation
4. The mean of the squared deviations

The balance condition of an AC bridge requires that

Options :

1. Minute current flows through detector
2. Power consumed is minimum
3. No current flows through the detector
4. Impedances are purely resistive

Wheatstone bridge is mainly used for measurement of

Options :

1. Low capacitance
2. Medium resistance
3. Low resistance
4. High resistance

In a Kelvin's double bridge two sets of readings are taken when measuring a low resistance, one with the current in one direction and the other with the direction of current reversed. This is done to

Options :

1. Eliminate the effect of constant resistance
2. Eliminate the effect of resistance of leads
3. Correct for changes in battery voltage
4. Eliminate the effect of thermoelectric EMFs

Question Number : 71 Question Id : 464662671 Question Type : MCQ

A Current Transformer (CT) is used to step down a primary current of 100 A to a standard anometer rated for 5 A, but at a certain load the CT secondary carries only 4.8 A with its rated burden. What is the ratio error (in percent) of the CT ?

Options :

1. 4%

2. 5%

3. 2%

4. 3%

Question Number : 72 Question Id : 464662672 Question Type : MCQ

The bandwidth of a CRO is from 0 – 20 MHz. The fastest rise time a sine wave can have to be accurately reproduced by the instrument is

Options :

1. 35 ns
2. 35 microsec
3. 17.5 ns
4. 0.175 microsec

The principle of operation of a Q-meter is based on

Options :

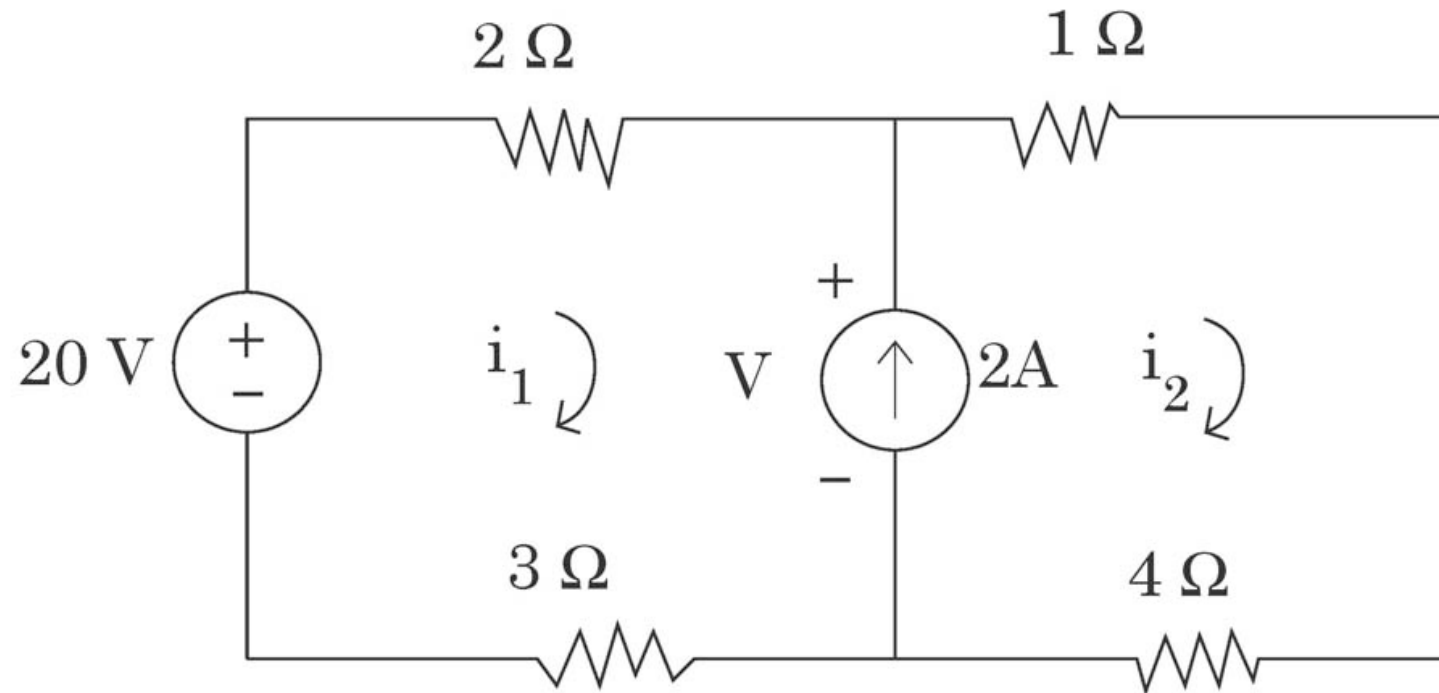
1. Parallel RLC resonance
2. Series RLC resonance
3. Wheatstone bridge balance
4. Transformer coupling

In measurement systems, which of the following static characteristics are desirable ?

Options :

1. Accuracy
2. Sensitivity
3. Reproducibility
4. All of the above

The current i_1 in the circuit is



Options :

1. 4 A
2. 3 A
3. 2 A

4. 1 A

Question Number : 76 Question Id : 464662676 Question Type : MCQ

The controlling torque in single phase power factor meter is provided by

Options :

1. Spring control

2. Gravity control

3. Stiffness of suspension

4. None of the above

In the two-wattmeter method of measuring power in a balanced three-phase load, one wattmeter reads zero, while the other reads positive power, the load power factor is

Options :

1. Unity
2. 0.5 Lagging
3. 0.0 (purely reactive)
4. 1.0 leading

If a homogeneous electrical conductor is subjected to temperature gradient, then proportional voltage gradient is generated along the conductor is known as

Options :

1. Peltier effect
2. Seebeck effect
3. Thompson effect
4. Boltzman effect

Question Number : 79 Question Id : 464662679 Question Type : MCQ

At what temperature do the Fahrenheit and Celsius scale coincide ?

Options :

1. 0
2. 20
3. 40
4. -40

Question Number : 80 Question Id : 464662680 Question Type : MCQ

A magnetic pickup type of turbine flow meter generates

Options :

1. Mechanical output
2. Magnetic output
3. Electrical digital output
4. Electrical analog output

Question Number : 81 Question Id : 464662681 Question Type : MCQ

The value of gauge factor for a semiconductor strain gauge used in practice can be approximately

Options :

1. 0.48
2. 2.05
3. 3.5
4. 150

Question Number : 82 Question Id : 464662682 Question Type : MCQ

A torsion bar of 30 mm diameter produces a twist of 1 minute when torque of 200 Nm is applied. Supposing the diameter of the torsion bar is reduced to 15 mm, what would be the angle of twist for the same torque ?

Options :

1. 2 minutes
2. 4 minutes
3. 0.125 minute
4. 8 minutes

DC-type capacitive transducer circuits are suitable only for

Options :

1. Static measurement
2. Thermal measurement
3. Dynamic measurement
4. Long-term measurement

A bolometer is a thermal device that changes its

Options :

1. Electrical resistance in response of its displacement
2. Electrical resistance in response of its thermal radiations
3. Electrical resistance in response of pressure acting on it
4. Electrical resistance in response of acceleration faced by it

A pressure applied to a bellows element produces a linear displacement of 2 mm. Suppose the number of bellows elements is made 3 times and thickness of the bellows element is halved, what would be the displacement of the element for the same applied pressure ?

Options :

1. 24 mm
2. 12 mm
3. 48 mm
4. 3 mm

Question Number : 86 Question Id : 464662686 Question Type : MCQ

A cantilever is used as a primary sensing element for measurement of force. The measured deflection is 0.5 mm for a particular set up. Suppose each of the dimensions i.e. length, width and thickness of the cantilever are doubled with the force and material of cantilever remaining the same, the deflection is

Options :

1. 1 mm
2. 2 mm
3. 0.251 mm
4. 0.125 mm

The resistance value of a thermistor is $5000\ \Omega$ at $25\ ^\circ\text{C}$ and its resistance temperature coefficient is $0.04/^\circ\text{C}$. A measurement with a lead resistance of $10\ \Omega$ will cause an error of

Options :

1. $0.05\ ^\circ\text{C}$
2. $0.1\ ^\circ\text{C}$
3. $0.02\ ^\circ\text{C}$
4. $0.01\ ^\circ\text{C}$

Question Number : 88 Question Id : 464662688 Question Type : MCQ

If the temperature of a radiating body is 1500°C , the wavelength at which the maximum radiant energy occurs _____.

Options :

1. $1\ \mu\text{m}$
2. $20.8\ \mu\text{m}$
3. $1.63\ \mu\text{m}$
4. $2.8\ \mu\text{m}$

Question Number : 89 Question Id : 464662689 Question Type : MCQ

While measuring liquid level with fibre optic transducers, the fibre rod or cable is cut in the form of

Options :

1. Sphere
2. Rectangle
3. Prism
4. None of the above

LVDT used for displacement measurement works on the principle of

Options :

1. Variable resistance
2. Variable self-induction
3. Variable mutual induction
4. Variable capacitance

Two main types of gyroscopic devices are

Options :

1. Active and Passive
2. Analog and Digital
3. Free gyro and Restrained gyro
4. Mechanical and Electrical

Which of the following flow meters is a volumetric flow meter and a direct method of measuring fluid flow ?

Options :

1. Turbine flow meter

2. Venturi meter

3. Orifice meter

4. Pitot tube meter

Light emission from an LED is modulated by

Options :

1. Voltage applied across the LED (diode)
2. Current passing through the LED
3. Illumination of the LED
4. All of the above

In a graded index fiber, the total reflected light takes a

Options :

1. Elliptical path
2. Circular path
3. Straight line path
4. Parabolic path

For the PIN photo diode which of the following statements are true ?

Options :

1.
An undopped intrinsic layer between two highly dopped regions wider than the intrinsic layer.
2.
A highly dopped region between two wider intrinsic layers.
3.
A highly dopped wider layer between two intrinsic layers.
4.
An undopped wider intrinsic layer between two highly dopped regions.

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The process of emission of light from a medium due to electric field applied across it is called

Options :

1. electrolysis
2. electromagnetism
3. electroluminescence
4. electroplating

The North American TDMA digital cellphone standard transmit at 24.3 Kiloband using DQPSK. What is the channel data rate ?

Options :

1. 48.6 Kb/s
2. 12.1 Kb/s
3. 24.3 Kb/s
4. 97.2 Kb/s

If the number of bits used to express the sample value are increased, which of the following will happen ?

Options :

1. Number of levels increase, data rate will also increase.
2. Number of levels increase, data rate will remain unchanged.
3. Number of levels increase, data rate decreases.
4. Number of level decrease, data rate increases exactly by a factor 2.

In frequency modulation if the carrier frequency is doubled, what is the effect on modulation index m_f ?

Options :

1. m_f is doubled
2. m_f is halved
3. m_f reaches maximum value
4. m_f not affected

Time division multiplexing is used when

Options :

1. the data to be transmitted is slow changing
2. the data to be transmitted has a small bandwidth
3. the data to be transmitted is slow changing and has a low bandwidth
4. the data to be transmitted is fast changing and has a high bandwidth